1. 2013Q21

Which of the following combinations shows the correct information about Eubacteria and Protista?

	Eubacteria	*Protista*
A.	bigger in size	smaller in size
B.	absence of cell wall	presence of cell wall
C.	presence of true nucleus	absence of true nucleus
D.	absence of mitochondria	presence of mitochondria

2. 2014Q15

Which of the organisms below belong to the domain Eukarya?

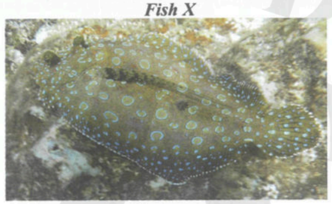
(1) Yeast
(2) Amoeba
(3) Mouse

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

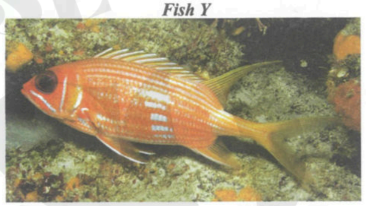
3. 2014Q17

Question 17 refers to the following photographs of two different fish:

Fish X



Fish Y



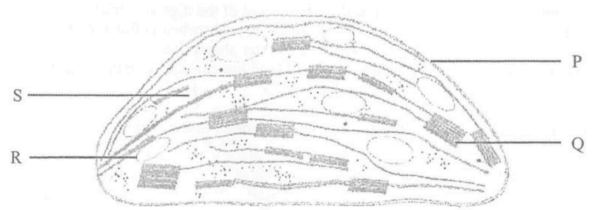
Using the dichotomous key below to identify the fish:

1a	Both eyes on the top of the head ……………………	2
1b	One eye on each side of the head ……………………	3
2a	Has long whip-like tail ………………………………	*Aetobatus narinari*
2b	Has short, blunt tail …………………………………	*Bothus mancus*
3a	Has spots on its body surface ………………………	4
3b	Does not have spots on its body surface………………	5
4a	Has chin whiskers ……………………………………	*Pseudupeneus maculatus*
4b	Does not have chin whiskers …………………………	*Sphoeroides spengleri*
5a	Has stripes on its body surface ………………………	*Holocentrus rufus*
5b	Does not have stripes on its body surface …………....	*Parapriacanthus guentheri*

	Fish X	*Fish Y*
A.	*Bothus mancus*	*Pseudupeneus maculatus*
B.	*Bothus mancus*	*Holocentrus rufus*
C.	*Aetobatus narinari*	*Pseudupeneus maculatus*
D.	*Aetobatus narinari*	*Holocentrus rufus*

4. 2015Q5

Question 5 refers to the schematic diagram below, which shows the structures of a chloroplast:



Which of the following kingdoms contains organisms that possess the above organelle?

(1) Eubacteria
(2) Protista
(3) Plantae

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

5. 2016Q15

Question 15 refers to the information below. Five new species of eubacteria were discovered in Antarctic ice core samples. The nucleotide sequences of the gene that codes for an essential protein of these new species were compared. The table below shows the number of nucleotide differences between the species:

	Number of nucleotide differences				
Species	P	Q	R	S	T
P	-	4	12	11	22
Q		-	12	11	19
R			-	4	22
S				-	22
T					-

Which of the following cell components can be found in these species?

A. nucleus
B. cell wall
C. chloroplast
D. mitochondrion

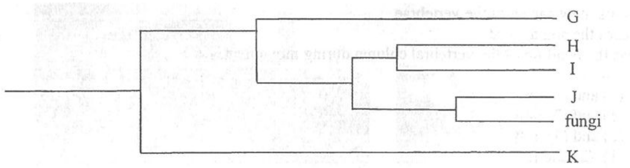
6. 2018Q9

Which of the following statements provides the best reason for classifying unicellular organisms into domain Bacteria and Archaea?

A. Archaea are more ancient than bacteria.
B. Archaea are smaller than bacteria.
C. The DNA sequences of archaea are distinct from those of bacteria.
D. The compositions of the cell wall and cell membrane of archaea are different from those of bacteria.

7. 2021Q28

Questions 28 and 29 refer to the diagram below, which shows an evolutionary tree demonstrating the phylogenetic relationship of the six kingdoms:



Which of the following combinations correctly shows the kingdoms represented by G, J and K in the evolutionary tree?

	G	*J*	*K*
A.	Archaebacteria	Animalia	Eubacteria
B.	Archaebacteria	Plantae	Eubacteria
C.	Eubacteria	Plantae	Protista
D.	Eubacteria	Animalia	Protista

8. 2021Q29

Which of the following pairs of organisms belongs to the same domain?

A. G and K
B. G and H
C. H and J
D. J and K

Bio - By topic - Topic 25 Biodiversity (2012-2024 version) by **DSERESO**

LQ

1. 2012Q4

The following key can be used for identifying organisms under the same phylum:

1a	Absence of eyes	--	2
1b	A pair of eyes	--	3
2a	Six legs	--	Class A
2b	More than six legs	--	Class B
3a	Six legs	--	Class C
3b	More than six legs	--	Class D

a) Using the above key, identify which class organism X shown in the photograph below belongs to. (1 mark)

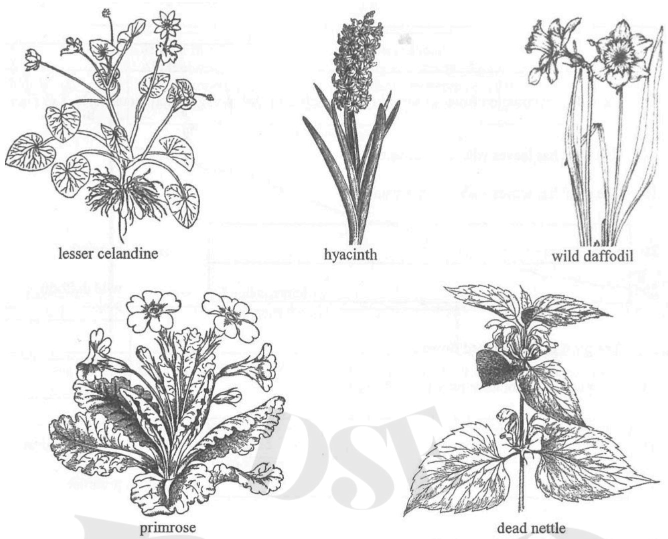
b) Suggest a characteristic of the habitat of organism X. Explain your answer. (2 marks)

c) A newly found organism Y has a pair of eyes and fewer than six legs. Although it is believed that this organism belongs to this phylum, it cannot be identified by using the above key. Explain why this problem occurs. (1 mark)

d) Suggest ***one*** way to collect more information which can be used for deciding whether organism Y belongs to this phylum. (2 marks)

2. 2016Q4

The following diagrams show the appearance of five flowering plants:



a) In the following table, put a '✓' in the appropriate boxes to show the features of each flowering plant. (2 marks)

	Leaves with parallel veins	Leaves with network veins	Single flower	Cluster of flowers	Other features
lesser celandine					heart-shaped leaves
hyacinth					funnel-like flowers
wild daffodil					trumpet-like flowers
primrose					club-shaped flowers
dead nettle					two-lipped flowers

b) Using the information from the table in (a), complete the following dichotomous key: (3 marks)

1a	The plant has leaves with parallel veins	-------------	2
1b	The plant has leaves with network veins	-------------	3
2a		-------------	hyacinth
2b		-------------	wild daffodil
3a	The plant has two-lipped flowers	-------------	
3b	The plant does not have two-lipped flowers	-------------	
4a		-------------	lesser celandine
4b		-------------	primrose

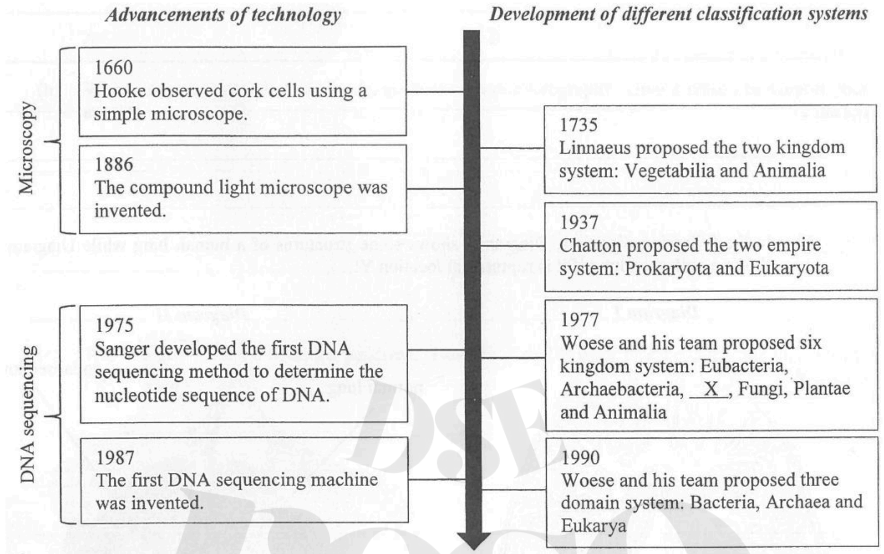
c) Study the following statement: (2 marks)

The dichotomous key shows that the lesser celandine and primrose have a closer evolutionary relationship.

Do you agree? Explain your answer.

3. 2017Q6

The chart below shows the timeline of some major advancements of technology and the development of different classification systems:



a) Name kingdom X in the six kingdom system proposed by Woese and his team in 1977. (1 mark)

b) How did the following technological advancements contribute to the development of different classification systems? (4 marks)

Microscopy:

DNA sequencing:

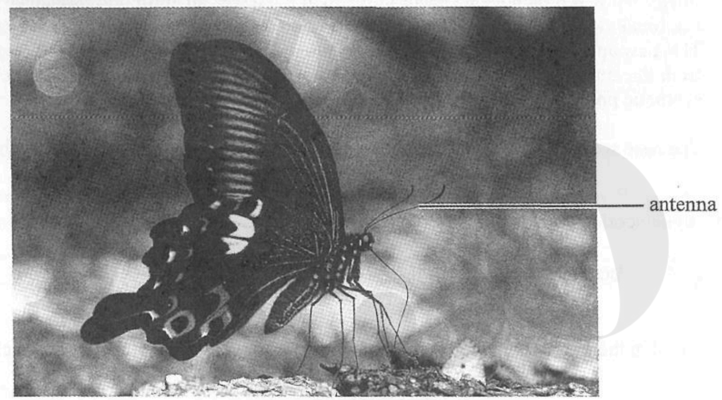
4. 2022Q10a

Habeneria is a genus of orchids with dull-coloured and scented flowers, which attract moths for pollination at night. In the group, *H. rhodocheila* is an exceptional species with reddish flowers which lack a detectable scent. It was found that no insects visited the flowers of *H. rhodocheila* at night for pollination while insect species A consumed nectar from the flowers in the daytime.

a) The following dichotomous key can be used to identify the group to which insect species A belongs:

1a	with wings	--	2
1b	without wings	--	Group P
2a	antennae longer than head	--	3
2b	antennae shorter than head	--	Group Q
3a	wings rest at their sides	--	Group R
3b	wings rest together upright	--	Group S

The photograph below shows the appearance of species A at rest. Using the above key, write down the sequence which leads to the identification. (1 mark)



Ans

2013Q21: D (54%)	2014Q15: D (35%)	2014Q17: B (81%)	2015Q5: C (74%)
2016Q15: B (51%)	2018Q9: C (41%)	2021Q28: A (12%)	2021Q29: C (60%)

2012Q4

4 (a) • Class A (1) (1)

(b) • the light intensity of the habitat is very low / the habitat is completely dark (1) (1)
• as organism X does not have eyes to survive in the habitat (1) (1)

(c) • the key is constructed based on the morphological characteristics of existing organism found (1) / not all the morphological characteristics of the phylum are listed in the key (1)

(d) • carry out a comparative study about the amino acid sequence of similar proteins / base sequence of DNA template / mRNA of similar proteins found in organism Y and other organisms in this phylum (1) (1)
• to establish the phylogenetic relationship between them (1) (1)

6 marks

2016Q4

4 (a)

	Leaves with parallel veins	Leaves with network veins	a single flower	a cluster of flowers	other features
lesser celandine		✓	✓		heart-shaped leaf
hyacinth	✓			✓	funnel-like flower
wild daffodil	✓		✓		trumpet-like flower
primrose		✓	✓		club-shaped leaf
dead nettle		✓		✓	two-lipped flower

(1) (1)

(2)

(b) 2a The plant has a cluster of / funnel-like flowers
2b The plant has a single / trumpet-like flower (1)

3a dead nettle
3b 4 (1)

4a The plant has heart-shaped leaves
4b The plant has club-shaped leaves (1)

(c) • this is incorrect (1)
• because a dichotomous key is used to identify organisms from a group based on the observable / morphological features which may not be related to their evolutionary / phylogenetic relationship (1) (2)

7 marks

2017Q6

6 (a) • Protista (1) (1)

(b) Microscopy:
- microscopy allows the observation of cellular structures (1)
- this distinguished prokaryotic cells from eukaryotic cells (1), giving rise to the basis of the two empire system proposed

DNA sequencing: (4)

(c)
- DNA sequencing determines the nucleotide sequence of the DNA of different organisms (1)
- so that the scientists can work out the phylogenetic relationship (1), giving rise to the basis of the three domains system

5 marks

2022Q10a

10 (a) • 1a → 2a → 3b (1), group S (1)